Object Detection: A Beginner’s Guide

A short learning document with clear concepts and examples for document loading and RAG experiments.

1. What is object detection?

Object detection is a computer-vision task that identifies objects in an image and estimates where each object is located. A detector commonly returns a class label, a bounding box, and a confidence score for each detection.

2. What does a detection contain?

- **Class label:** the predicted category, such as person, car, dog, or bicycle.
- **Bounding box:** a rectangle around the object, often represented by coordinates such as (x_min, y_min, x_max, y_max).
- **Confidence score:** a model-estimated score indicating how strongly the prediction matches a class.

3. Example

Suppose a street image contains a car and two people. An object-detection model might return three detections: “car” with a box around the vehicle, “person” with a box around the first person, and “person” with a box around the second person. The confidence scores help an application decide which predictions to keep using a chosen threshold.

4. Object detection versus image classification

Image classification assigns one or more labels to an entire image, such as “street scene.” Object detection goes further: it identifies individual objects and locates each one. Image segmentation is different again; it can assign labels at the pixel level to outline object shapes more precisely.

5. Common stages in an object-detection workflow

- Collect or load an image.
- Preprocess the image to match the model’s expected size and format.
- Run a trained detection model.
- Interpret the predicted labels, boxes, and confidence scores.
- Optionally filter low-confidence predictions and draw the boxes on the image.

6. Where embeddings fit in

Some vision models can convert images or image regions into embeddings. These vectors can be used to compare visual similarity or retrieve related examples. This is not the same as standard object detection: an embedding alone does not necessarily provide object labels or bounding boxes. A RAG system can retrieve text about object detection, while a computer-vision model performs detection on an image.

7. Example questions for testing a RAG pipeline

- What is object detection?
- What information does a bounding box represent?
- How is object detection different from image classification?
- What does a confidence score mean?

- Are image embeddings the same thing as object-detection outputs?

8. Key takeaway

Object detection locates and classifies objects in images. A text-based RAG pipeline can retrieve explanations about this topic from a PDF, but it will not detect objects in a new image unless you also integrate a suitable vision model.

This document is educational sample content, not a dataset or a benchmark for measuring detection accuracy.